

BEFORE

In the 15th century the Portuguese explored the west coast of Africa.

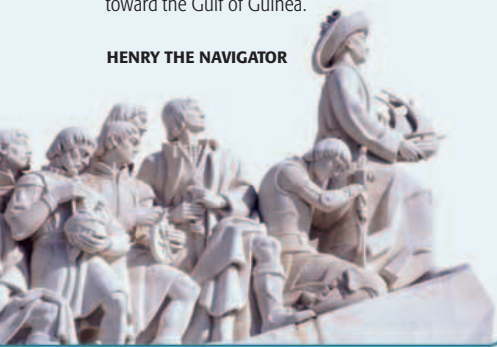
HENRY THE NAVIGATOR

Prince Henry of Portugal captured Muslim Ceuta on the north coast of Africa in 1415 and was eager to extend his knowledge of the continent. The next year he set up a **navigation school at Sagres** to provide Portuguese sailors with a center for exploration. Here they learned the **necessary skills of navigation, seamanship, and cartography** to enable them to venture overseas. Each year he sponsored a voyage south to explore the African coast, each voyage sailing farther south than its predecessor.

THE FIRST SUCCESSES

One of Henry's navigators discovered the Madeira Islands in 1418–20 and the Azores in 1427–31. **As the navigators returned home with gold and slaves**, African exploration became very popular. By the time of his death in 1460, the Portuguese had discovered the Senegal River, rounded Cape Verde, and sailed east toward the Gulf of Guinea.

HENRY THE NAVIGATOR



Expedition routes

This map shows the major European ocean voyages of the 15th and 16th centuries. Each line color corresponds to the country of the explorer, whose name and expedition date appear above the line. Arrows indicate the direction of travel, and major ports or stopovers are indicated with white dots.

“May the ocean always be as calm and benevolent as it is today. In this hope I name it the **Pacific Ocean**.”

FERDINAND MAGELLAN ON FIRST ENTERING THE PACIFIC OCEAN, NOVEMBER 27, 1520

Voyages of Discovery

In little over a century, European navigators left their continent and sailed the world, opening up new sea routes to India and the east. They discovered a continent previously unknown to them, and began a process that eventually resulted in the total European colonial and economic domination of the world.

PORTUGUESE EXPLORER (1480–1521)

FERDINAND MAGELLAN

Magellan was a Portuguese soldier and adventurer who had taken part in four expeditions to India and Malaya. He quarreled with the Portuguese king, Manuel I and left the country in 1514 to enter the service of Spain. In 1519 he proposed a voyage west to the Spice Islands, which promised great wealth to Spain if successful. Magellan set out in 1519 with five ships and about 260 men. He sailed south across the Atlantic, and in November 1520 through the straits that now bear his name and into the Pacific. He then crossed the ocean, the first European to do so, reaching the Philippines in April 1521, where he was killed in a local war. His deputy, Juan Sebastian del Cano, took charge, eventually returning to Spain in September 1522 with only one ship and 17 members of the crew.



At the start of the 15th century, European knowledge of the world was surprisingly limited. Sailors used world maps based on the cartography of Ptolemy, a Greek geographer who had died 13 centuries earlier, in 168 CE. Europe, the Mediterranean, and western Asia were reasonably well mapped, but Africa was vague in shape, as no one knew how far south it stretched, and the Americas were missing altogether. Poor ship design that restricted oceangoing voyages and primitive navigation instruments kept ships close to the coast. At best, navigators used dead reckoning—intelligent guesswork based on speed through the water, winds, and currents—to assess their location.

The development of the oceangoing caravel (see right) by the Portuguese and new navigation instruments transformed this situation. The magnetic compass, the astrolabe, which measured the height of the Sun at noon, and the cross-staff and quadrant, which measured the height of a star, all helped navigators determine their latitude, or how far north or south they were. Lack of accurate marine chronometers (see p.282) meant that longitude—distance east or west—was not accurately calculated until the mid-18th century.

New technology, new worlds

Armed with this new maritime technology, the Portuguese tentatively explored the coast of Africa. Diogo Cão sailed around the Gulf of Guinea and then headed south, exploring the Congo River before making his final landfall at Cape Cross in what is now Namibia in 1486. Two years later, Bartolomeu Dias rounded the Cape of Good Hope and sailed into the Indian Ocean, while in 1498 Vasco da Gama crossed the Indian Ocean to Calicut in southern India. These expeditions